

Josephus Queries

Time limit: 1.00 s

Memory limit: 512 MB

Consider a game where there are n children (numbered $1, 2, \dots, n$) in a circle. During the game, every second child is removed from the circle, until there are no children left.

Your task is to process q queries of the form: "when there are n children, who is the k th child that will be removed?"

Input

The first input line has an integer q : the number of queries.

After this, there are q lines that describe the queries. Each line has two integers n and k : the number of children and the position of the child.

Output

Print q integers: the answer for each query.

Constraints

- $1 \leq q \leq 10^5$
- $1 \leq k \leq n \leq 10^9$

Example

Input	Output
4	2
7 1	6
7 3	1
2 2	1107
1337 1313	